

The role of competition and macroprudential regulation on bank liquidity creation: The case of Indonesia

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Abstract

This paper provides insights into the role of competition and macro prudential regulation on improving Indonesia banking liquidity creation post global financial crises 2008/09. The results suggest that macro prudential measure of reserve requirement linked to loan to deposit ratio is effective enough to increase liquidity creation. Transparency of prime lending rate is effective to foster stronger competition among banks. Stronger competition somehow leads to lower liquidity creation due to banking risk averse behaviour, particularly among small banks. Therefore, consolidation among small banks may be a good idea to have greater market power that enables them to create more liquidity.

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I. Introduction

Studies have been dedicated to examine the role of banking competition on banking stability because of its implication on policy decision making. Existing empirical works focus on the role of competition on banks' risk taking behaviour with credit risk, liquidity risk, insolvency risk, and systemic risk as measures for various risks dimension.

However, banking stability is not only related to ability of banking to withstand various shocks or risks, but is also related to the ability of banking to keep its primary function of funds intermediation effective (ECB, 2015). This primary role of banking is also known as liquidity creation. Banks create liquidity on their balance sheets by financing relatively illiquid assets or investments with relatively liquid liabilities (Diamond & Dybvig, 1983) or on their off balance sheets through loan commitments and other liquidity claims (Kashyap *et al.*, 2002). These intermediation activities may expose banks to *asset-liability maturity mismatch* risk. Diamond & Rajan (2000, 2001, and 2002) argue that bank failures are more likely to occur when the level of liquidity creation is high. The 2008 global financial crisis has taught us a valuable lesson that excessive lending without prudent behaviour of the lenders would have resulted in a crisis that leads to deep recession in many countries. Therefore, regulators need to ensure the liquidity created by banks is carried out with prudent behaviour.

Following the financial global crises in 2008, Indonesia economy suffers credit crunch. Credits have only grown around 10% by the end of 2009 which is much lower than the 30% growth in 2008. Loan to deposit ratio (LDR) indicated low banking intermediation as it declined from 80.4% in 2008 to the level of 74.5% in 2009. This credit crunch coincided with sharp declining in economy growth from 6.01% in 2008 to only 4.55% in 2009. As Indonesia is a bank based economy, Bank Indonesia (BI) issued some initiatives to foster banks' investment activities that are expected to have positive impact on real domestic output. From monetary side, BI started with dramatic cut off of the policy interest rates from 9.25% in 2008 to 6.50% in 2009 and

continued to 5.75% in 2012. This effort was complemented with two initiatives in banking regulation in 2011 indirectly and directly aimed at improving banking liquidity creation. First, BI released regulation of transparency of prime lending rate in first quarter of 2011. The objective of this initiative is to boost credit price competitiveness up among banks in attempt to lower loans interest rates charged to costumers. Banks are required to display their prime lending rate publicly so that potential debtors may compare and choose cheaper rates. Second, to encourage banks to create more liquidity while ensuring that expansion is also carried out within prudence corridors, BI issued a macro prudential regulation of reserve requirement linked to loan to deposit ratio (LDR). This new measure charges penalty in terms of higher reserve requirement for banks with LDRs are out of tolerable range of 78%-100%. This regulation effectively took place from first quarter of 2011 to 2013.

This paper is aimed at evaluating whether or not these two initiatives are effective to meet the objective of boosting banking competitiveness and liquidity creation. As many researches have been dedicated to evaluate the role of banking competition on banking stability but only a few on banking intermediation role, my research will fill the gap by sheeding insights into the role of banking competition and regulations on banks liquidity creation. More specifically, the objective of this study is to examine hypothesis on whether or not stronger competition in the banking industry and macro prudential regulation are significant to encourage banks to create more liquidity.

The findings could be an important feedback for policy evaluation and future designing. To my knowledge, this is the first paper that investigates the role of macro prudential regulation on bank's intermediation role in conjunction with competition by utilizing a novel indicator of Liquidity Creation (LC) introduced by Berger and Bouwman (2009). The LC indicator by construction does not only represent banks' willingness to channel loans but also to issue deposits. It measures both banks' intermediation level and risk taking behavior (liquidity risk). Higher value of LC not only represents higher liquidity creation but also higher potency of liquidity risk.

This study utilizes Indonesia data set. Indonesia is a unique case as it is the largest bank based economy in south east Asia. Its banking sector has dealt with two major external crises, the 1997/98 Asian financial crises and the 2008/09 global financial crises as well as some domestic turmoils. This vast experience contributes on how both regulator and bankers perceived risks, which is reflected in adopted policies and business approaches. Therefore, this research is expected to set a new strand of literature in field of banking competition-intermediation.

I utilize two stages instrumental variable (IV) analysis to evaluate the role of competition and macro prudential measure on banking liquidity creation. In the first stage, I examine the role of regulation to induce competition by instrumenting bank level competition measure of Lerner index with regulation in transparency of prime lending rate. To my knowledge, this paper is among few researches that identify competition with regulation. In the second stage, I evaluate the role of predicted competition from the first stage as well as macro prudential regulation of reserve requirement linked to LDR on liquidity creation. In general the results indicate that macro prudential regulation of reserve requirement linked to LDR is effective enough to increase banking liquidity creation. Regulation of the transparency in prime lending rate is also effective to foster stronger price competition among banks. However, some evidences show this stonger competition leads to lower liquidity creation due to banks risk averse behaviour. These main results are relatively robust as I also do placebo regressions by shifting the implementation dates of the two regulations up to 3 years earlier as well as introduce two alternative measures of liquidity creation.

That stronger competition contributes to lower liquidity creation can be also explained by evaluating the effectiveness of this regulation from price dimension. I examine the hypotheses if stronger competition due to regulation in transparency of prime lending rate leads to significantly lower banks loans interest rates, which is the objective of the regulation. Therefore, in the second stage analysis I regress loans interest rates on predicted Lerner from the first stage of IV analysis including same control variables.

The result indicates that stronger competition due to implementation of transparency in prime lending rate encourage banks to charge customers higher loan interest rates. This may be caused by banks attempt to maintain their profit level as stronger competition seems to erode banks profit, which is inline with the competition-fragility theorem. Moreover, the result also reveals that higher credit risk encourages banks to increase their loan interest rates which strengthen the tendency of banks' risk averse behaviour. Under assumption that higher loans rates will reduce demand for loans leading to lower liquidity creation, *ceteris paribus*, this will support the conclusion drawn from the main analysis that stronger competition actually leads to lower liquidity creation that may be attributed to banking risk averse behaviour.

This paper is structured into five parts. The first part describes the background that motivates this study as well as its contribution. The second part explains relevant literatures, while data, variables, hypotheses, model and identification strategy as well as estimation technique are discussed in detail in part three. I present the main results, robustness check, alternative explanation as well as future extension in part four, and conclude it along with its policy implications in part five.

II. Literature Review

This study is built on two strands of literatures as I will briefly explain.

Competition and liquidity creation

Existing empirical works have been focussed on the link of competition on banking stability with various risks dimension as measures for banking stability. The results are still ambiguous. The works of Allen & Gale (2004), Hellman et al. (2000), Cordella & Levi-Yeyati (2002), Beck, *et al.* (2003), Repullo (2004), as well as Jimenez, *et al.* (2007) support charter value paradigm where competition leads to banking fragility (Keeley, 1990). Higher degree of competition may lead to lower loan rates and higher deposit rates implying lower profit margins for banks. This condition erodes banks' franchise value and encourages banks to take more risks.

On the contrary, the works of Boyd, *et al.* (2006), Shaeck, *et al.* (2009), Poghosyan (2012), and Anginer, *et al.* (2012) support the competition-stability hypothesis. They argue that higher degree of competition will improve price efficiency as banks charge lower interest rates for loans. This will benefit both banks and borrowers as lower loan repayment will reduce probability of loan default, therefore improving the stability of banking. The last empirical study that portrays these conflicting results is done by Almarzoqi, *et al.* (2015). They utilize annual data from 1999-2013 on 367 banks in Middle East and North Africa countries and find that price competition (Lerner index) has a positive effect on bank liquidity, but a negative impact on bank credit quality and solvency.

However, risks are only one dimension of banking stability. Another important dimension of banking stability is the ability of banking to keep its primary function of creating liquidity or funds intermediation for the economy effective (ECB, 2015).

Despite the importance of the banking role in funds intermediation, very few empirical works have been carried out to examine the link between banking competition and its primary role of liquidity creation. Horvath, *et al.* (2016) examine impact of banking competition on liquidity creation. Authors use Czech banks quarterly data from 2002-2010 and find that enhanced competition (lower market power) reduces banks' liquidity creation, giving support to competition-fragility theorem.

Macro prudential regulation and liquidity creation

This is a relatively new strand of literature. There are still not many papers discuss on the link between macro prudential regulation and liquidity creation. Macro prudential regulation gains immense attention from financial regulators following the devastated impact of global financial crisis in 2008/09. This regulation aimed at providing incentives or disincentives to promote prudent behaviour practise in financial industry to minimize the occurrence of financial fluctuations that leads to systemic risk. Distinguin, *et al.* (2013) model the role of bank regulatory capital on

liquidity creation and find that higher tier 1 regulatory capital ratio leads to lower liquidity creation for European banks. Adrian & Boyachenko (2012) argue that regulators are actually exposed to the trade-off between tighter regulation that may reduce banks' risk taking incentives thus vulnerability and more expensive credit price. Given liquid liabilities, this condition imply for lower banks' liquidity creation. Berger, *et al.* (2016) confirm this trade off between having a more stable banking system and lower liquidity creation. They employ Germany banking annual data over the period of 1999-2009 and find that regulatory interventions (in terms of restrictions and prohibitions of deposit taking, restrictions and prohibitions of profit distribution, restructuring of business activities and limitation on the scope of managerial decisions) reduce not only banks' risk taking incentives but also liquidity creation.

III. Data and Methodology

III.1 Data & variables

I utilize quarterly data for 96 out of 119 Indonesia commercial banks over the 2004-2014 period. This sample banks cover around 90% of Indonesia total banking asset. I exclude Syariah banks as they have different business characteristics from conventional banks. I also exclude the foreign branch banks as these banks get both capital and liquidity back up from parent company abroad. Source of data are monthly report of banking and macroeconomic statistics from Bank Indonesia and Indonesia Financial Service Authority (OJK).

To ensure extreme data do not affect the estimation results, I exclude observations with capital adequacy ratio (CAR) exceed 100% and lower than 8% as regulator requires banks to maintain CAR at minimum level of 8%. Other variables are winsorized either at the level of 5% and 95% or 1% and 99%, depending on the variability level of data.

Data for liquidity creation and bank competition are constructed based on the following explanation.

Liquidity creation measure

Following Berger & Bouwman (2009), I construct a liquidity creation measure (LC) utilizing Indonesia banking data. However, given Indonesia banking focuses on traditional intermediary business with limited exposure to derivative activities rather than acts as investment banking, I adopt the narrow version of category based LC that utilizes only on balance sheet items and excludes the derivative and off-balance sheet activities. I also adjust the definition of commercial loan (includes working capital and long term investment loan) and consumer loans following regulator's standards.

The LC construction involves three stages. First, bank activities are classified as illiquid, semi-liquid, and liquid based on ease, cost, and length of time for customers to achieve liquid funds from the bank (for assets side) or time for banks to meet depositors' liquidity demands (for liabilities side). Second, Berger & Bouwman (2009) assign positive weights on illiquid assets and liquid liabilities, negative weights on liquid assets and illiquid liabilities, and zero weights are for semiliquid assets and liabilities. They use 1/2 and -1/2 for positive and negative weights respectively, based on assumption that around half of the total amount of liquidity created is attributable to the source or use of funds. Stage 1 and 2 are summarized in Table 1 and 2.

Tabel 1 *Liquidity Creation – Assets side*

<i>Assets</i>	<i>Liquidity level</i>	<i>Weights</i>
<i>Cash</i>	<i>Liquid</i>	<i>-0.5</i>
<i>Short-term marketable assets</i>	<i>Liquid</i>	<i>-0.5</i>
<i>Long-term marketable assets</i>	<i>Semi liquid</i>	<i>0</i>
<i>Interbank assets</i>	<i>Semi liquid</i>	<i>0</i>
<i>Consumer loans</i>	<i>Semi liquid</i>	<i>0</i>
<i>Other loans</i>	<i>Semi liquid</i>	<i>0</i>
<i>Commercial loans (working capital and investment loans)</i>	<i>Illiquid</i>	<i>0.5</i>

<i>Assets</i>	<i>Liquidity level</i>	<i>Weights</i>
<i>Fixed assets</i>	<i>Illiquid</i>	<i>0.5</i>
<i>Other assets</i>	<i>Illiquid</i>	<i>0.5</i>

Source: Berger & Bouwman (2009), Distinguin, *et al* (2013), modified

Tabel 2 Liquidity Creation – Liabilities side

<i>Liabilities</i>	<i>Liquidity level</i>	<i>Weights</i>
<i>Demand deposits</i>	<i>Liquid</i>	<i>0.5</i>
<i>Saving deposits</i>	<i>Liquid</i>	<i>0.5</i>
<i>Short-term borrowings</i>	<i>Liquid</i>	<i>0.5</i>
<i>Other short-term borrowings</i>	<i>Liquid</i>	<i>0.5</i>
<i>Time deposits</i>	<i>Semi liquid</i>	<i>0</i>
<i>Long-term borrowings</i>	<i>Semi liquid</i>	<i>0</i>
<i>Other long-term liabilities</i>	<i>Semi liquid</i>	<i>0</i>
<i>Total Equity</i>	<i>Illiquid</i>	<i>-0.5</i>

Source: Berger & Bouwman (2009), Distinguin, *et al* (2013), modified

In final stage, LC in terms of a ratio is calculated as weighted sum of all components of asset and liability as depicted in Table 1 and 2, divided by total asset:

$$LC = \frac{(0.5illiquid\ asset - 0.5liquid\ asset) + (0.5liquid\ liability - 0.5illiquid\ liability)}{Total\ asset} \quad (1)$$

Higher value of LC indicates more liquidity created by banks as well as higher potency of liquidity risk.

Competition measure

Lerner index is a measure of bank's competitiveness stemming from pricing power that varies at the bank level. It measures both the impact of pricing power on the asset and funding side of the bank (Beck et al, 2013), which fits well with the

definition of liquidity creation. It is the only indicator that measures price competitiveness at individual bank level.

Lerner index is defined as the ratio of the profit or difference between price and marginal cost to price.

Following the methodology adopted by Demirguc-Kunt & Martinez-Peria (2010) and Anginer, *et al.* (2012), I start by estimating bank's cost function which is specified as:

$$\begin{aligned} \text{Log}(C_{it}) = & \gamma_0 + \gamma_1 \log(A_{it}) + \gamma_2 0.5 (\log(A_{it}))^2 + \gamma_3 \log(E_{1,it}) + \gamma_4 \log(E_{2,it}) + \\ & \gamma_5 \log(E_{3,it}) + \gamma_6 \log(A_{it}) \log(E_{1,it}) + \gamma_7 \log(A_{it}) \log(E_{2,it}) + \gamma_8 \log(A_{it}) \log(E_{3,it}) + \\ & \gamma_9 (\log(E_{1,it}))^2 + \gamma_{10} (\log(E_{2,it}))^2 + \gamma_{11} (\log(E_{3,it}))^2 + \gamma_{12} \log(E_{1,it}) \log(E_{2,it}) + \\ & \gamma_{13} \log(E_{1,it}) \log(E_{3,it}) + \gamma_{14} \log(E_{2,it}) \log(E_{3,it}) + \varepsilon_{it} \end{aligned} \quad (2)$$

The cost function models one output (total assets A_{it}) and three input prices (price of borrowed fund $E_{1,it}$, price of labor $E_{2,it}$, and price of physical capital $E_{3,it}$) to explain bank's total cost C_{it} .

P_{it} = price of assets = ratio of total revenue to total assets bank i at year t

C_{it} = total costs of bank i at year t

A_{it} = total assets of bank i at year t

$E_{1,it}$ = ratio of interest expenses to total deposits of bank i at year t

$E_{2,it}$ = ratio of personnel expenses to total assets of bank i at year t

$E_{3,it}$ = ratio of (general and administrative expenses, depreciation, and other operating expenses) to total assets of bank i at year t

I estimate the translog cost function with least square method for each year in order to get banks' dynamic cost function that vary across observed period of 2004-2014.

The estimated coefficients of the cost function are then used to compute *marginal cost of a bank in each period*:

$$MC_{it} = (C_{it} / A_{it}) * (\hat{\gamma}_1 + \hat{\gamma}_2 \log(A_{it}) + \hat{\gamma}_6 \log(E_{1,it}) + \hat{\gamma}_7 \log(E_{2,it}) + \hat{\gamma}_8 \log(E_{3,it})) \quad (3)$$

By defining price of asset P_{it} as a ratio of total revenue to total assets, the **Lerner Index** of bank i at period t is calculated as:

$$\text{Lerner}_{it} = (P_{it} - MC_{it}) / P_{it} \quad (4)$$

Higher Lerner value is associated with more bank's market power or lower bank's competitiveness.

Explanatory Variables

I employ eight explanatory variables to model the liquidity creation. The first six ones are the control variables while the last two variables are represented the two regulations of BI.

- **CAR** is a proxy for bank resilience as well as potency to support bank's liquidity creation. It is calculated as ratio of Equity to Risk Weighted Asset
- **NPL** is a proxy for bank credit risk. It is defined as ratio of non performing loan to total loan
- **Earning volatility** is another proxy for bank risk. It is calculated as the standard deviation of bank's return on assets ratio over the previous twelve quarters.
- **Ln Asset** is a proxy for bank's size. It is measured as the natural log of total asset in trillion IDR
- **Lerner**: is a proxy for bank's market power or pricing competitiveness.
- **GDP** is a proxy for economy development or business cycle, measured as year on year growth of real GDP.
- **BI** is a monetary policy interest rate indicates monetary policy stance as responses for domestic price level condition
- **d_RRLDR**: dummy of macro prudential regulation of reserve requirement linked to bank's loan to deposit ratio. Code 1 for periods of Q1 2011 - Q4 2013 and 0 otherwise
- **d_PLR**: dummy of regulation transparency of prime lending rate. Code 1 for periods starting from Q1 2011 to last sample period and 0 otherwise

Hypothesis on relationship between explanatory variables and liquidity creation

Horvath, *et al.* (2016) postulate two possible theorems on the link between competition and liquidity creation, the competition-fragility (lower liquidity creation) and the competition-price channel (higher liquidity creation).

In the first theorem of competition-fragility, higher competition reduces banks' profits which act as buffer against adverse shocks. Lower banks' buffer encourages banks to behave more prudently by limiting both the volume of loans granted (by raising loan interest rates to lower the likelihood of loan losses) and the volume of deposit issued (by lowering deposit rates) to reduce the likelihood of bank runs. As the consequence, banks' liquidity creation decreases.

In the latter theorem they argue that intense competition influences banking pricing policies, leading to diminished loan rates and increased deposit rates. As the consequence, demand for both loans and deposits rises or banks experience higher liquidity creation. Meanwhile, in a less competitive environment, banks with high market power are more likely to increase its own charter value by raising lending interest rates and lowering deposit rates, therefore reduces liquidity creation.

There are two possible relationships between banks' capital and liquidity creation. Berger & Bowman (2009) explain two different channels through which higher banks' capital may lead to lower liquidity creation, the "financial fragility structure" and the "the crowding out of deposit. Supported by the work of Diamond & Rajan (2000), the *financial fragility structure* theorem argues that higher capital is associated with lower incentives to monitor debtors' performance, leading to lower extended loans that yields lower liquidity creation. Gorton & Winton (2000) conclude that higher capital ratio may cause deposit *crowding out*. Banks with higher solvency level will attempt to reduce cost of borrowed funds by offering lower deposit interest rates, thus discourage investors to save their money in banks. As deposits grow less and so do the channelled loans, leading to lower liquidity creation. Horvath, *et al* (2016) utilize Czech banking data over the period of 2002-2010 and find there is a *trade-off between maintaining high capital ratio and creating lower liquidity*.

On the contrary, Repullo (2004) and von Thadden (2004) conclude that higher capital ratio is an incentive for banks to increase their liquidity creation as well as it

provides banks more buffer to absorb unanticipated loss that may emerge from the risks associated with the activities.

I employ two risk measures to control for bank risks as well as to separate the role of CAR in supporting the liquidity creation of banks from the role of capital in absorbing negative consequences of banks' risky activities (Berger&Bouwman, 2009). The first measure is the earning volatility ROA_vol. The second one is the ratio of non performing loan to total loan which is a measure of bank credit risk. Their impact on liquidity creation may depend on banks' risk appetite. Positive impact implying that higher uncertainty of earning or risk may induce banks to be more aggressive in expanding liquidity creation in an attempt to secure more profit to compensate for the potential loss. The negative impact confirming that higher risk encourages banks to be more cautious in doing liquidity creation.

I employ year on year GDP growth as a proxy for business cycle that affects banks' investment plan thus willingness to supply loan and to issue deposits. On expansion phase, the potency of loan demand is higher, leading to higher bank's liquidity creation. Thus, we expect positive relationship between GDP and LC.

I utilize policy interest rate of BI as a proxy for monetary stance of the central bank. As banks try to maintain their profits, increasing BI will be responded by increasing deposits and loans interest rates. As the consequence, banks' volume of deposit will grow higher while volume of credit will grow less, leading to higher liquid liability and smaller illiquid asset thus lower liquidity creation. Therefore, there will be negative relationship between BI and LC.

Implementation of macro prudential reserve requirement linked to loan to deposit ratio is expected to improve bank's liquidity creation within prudent corridor, therefore it is hypothesized d_RRLDR will have positive relationship with LC.

Table 3 summarized the hypotheses explained above.

Table 3 Summary of hypotheses

Explanatory Variable	Expected relationship with LC
Lerner	(-)/(+)

Explanatory Variable	Expected relationship with LC
CAR	(-)/(+)
ROA_vol	(-)/(+)
NPL	(-)/(+)
GDP	(+)
BI	(-)
d_RRLDR	(+)

III.2 Methodology

III.2.1 Identification strategy

There is potential causal link between liquidity creation and bank's capital ratio. Berger & Bouwman (2009) argue that this endogeneity problem stems from the fact that in practice banks simultaneously determine capital ratio and liquidity creation.

There is also potential endogeneity between LC and Lerner. High LC may also indicate bank's higher liquidity risk that affects their risk behaviour. Banks with high LC may try to attract new deposits aggressively by offering higher prices for their products. This will affect competitiveness in pricing of borrowed funds between banks. To overcome this endogeneity problem, I regress LC on lagged values of Lerner as well as two stages Instrumental Variable (IV) regression. Employing two stages IV regression also helps to overcome potential omitted variables bias that may arise, as some explanatory variables need to be excluded due to serious multicollinearity with other variables.

Taking into account that transparency of prime lending rate is directly aimed at promoting pricing competitiveness among banks while macro prudential measure of reserve requirement linked to loan to deposit ratio is aimed directly at affecting banks liquidity creation, I do the two stages IV analysis. In the first stage, I identify Lerner with regulation transparency of prime lending rate d_PLR as an IV. The use of d_PLR as an IV for Lerner must satisfy two important conditions, (i) it should statistically significant predict the actual Lerner and (ii) it should not have a direct

effect to liquidity creation. By design, requirement (ii) has been fulfilled as regulation of prime lending rate is directly aimed at promoting stronger price competitiveness among banks or to have negative relationship with Lerner index with indirect expectation on increasing bank liquidity creation. We need to check the significance of estimated coefficient for d_PLR to evaluate if condition (i) has been satisfied.

In the second stage, I estimate LC on predicted value of Lerner from the 1st stage as well as d_RRLDR that represents macro prudential measure of reserve requirement linked to LDR.

III.2.2 Regression Framework

Based on the identification strategy, I do baseline and two stages IV regressions. I estimate the models by employing fixed effect leastsquare (FELS) method with robust standard errors clustered at the bank level to account for heteroscedasticity and within-panel serial correlation. I include bank fixed effects to control unobserved bank time invariant heterogeneity and time fixed effect to control time variant of other macro or micro factors development that may affect all banks.

Baseline regression

To alleviate potential endogeneity problem between liquidity creation LC and bank's control variables, I regress LC on lagged values of the control variables. Since macroeconomic and monetary policy may affect bank's liquidity creation after sometimes, I also utilize lagged values of GDP and BI.

$$LC_{it} = constant + \alpha_i + \beta_1 Lerner_{t-k} + \beta_2 Bankcontrol_{i,t-k} + \beta_3 GDP_{t-k} + \beta_4 BI_{t-k} + \beta_5 d_RRLDR_t + v_t + \epsilon_{it} \quad (5)$$

I do not include d_PLR as a control variable because this variable exposes multicollinearity problem with d_RRLDR given the two regulations were issued in the same period and lasted almost in the same time duration

Two stages IV regression

In the first stage, Lerner is instrumented by prime lending rate regulation. I exclude d_RRLDR as a control variable due to serious multicollinearity problem with d_PLR given both regulations were issued at the same time. Moreover, I include time fixed dummy v_t to account for any other factors that may have impacted banks competition.

1st stage

$$Lerner_{it} = constant + \alpha_{1i} + \beta_{11}d_PLR_t + \beta_{12} \ln Asset_{t-k} + \beta_{13}Bankcontrol_{i,t-k} + \beta_{14}GDP_{t-k} + \beta_{15}BI_{t-k} + v_t + \epsilon_{it} \quad (6)$$

In the second stage, I regress LC on Lerner values ($Lerner_hat$) predicted from the first stage. I include $\ln Asset$ to control for bank size as there is no serious multicollinearity problem between this variable and other explanatory variables employed in the first stage regression. Larger banks are expected to have more market power than the smaller ones, therefore I hypothesize positive relationship between $\ln Asset$ and Lerner.

2nd stage

$$LC_{it} = constant + \alpha_{2i} + \beta_{21}Lerner_hat_{t-k} + \beta_{22}Bankcontrol_{i,t-k} + \beta_{23}GDP_{t-k} + \beta_{24}BI_{t-k} + \beta_{25}d_RRLDR_t + v_t + \epsilon_{it} \quad (7)$$

To account for potential problem that may arise due to excluding of $\ln Asset$ as a control variable for bank size as well as to check if results may be driven by bank size, I do separate regressions for the group of 20 large banks and 76 small banks in addition to regressions that utilizing full sample of 96 banks. A bank is categorized into the large group as per regulatory standards that sorted banks based on its asset size, business complexity, and interconnectedness with other banks or financial institutions. The 20 large banks share around two thirds of total industry asset.

IV. Results

IV.1 Descriptive statistics and coefficient correlation

Table 4 below displays the descriptive statistics of main variables utilized in this study.

Table 4. Descriptive statistics

	CAR (%)	LC (%)	LDR(%)	NPL (%)	Lerner (%)	Ln Asset	ROA_vol	GDP (%)	BI (%)
mean	22.68	40.45	78.24	2.63	62.23	1.50	6.54	5.73	7.79
median	18.33	42.00	79.84	1.86	86.47	1.39	1.68	5.87	7.44
min	8.22	12.00	38.38	0.00	-51.75	-3.91	0.04	4.10	5.75
max	99.22	63.00	125.79	51.33	99.68	6.66	18.06	7.16	12.75
sd	13.80	14.35	17.92	4.13	49.74	1.85	7.76	0.77	1.80

Table 5. Spearman coefficient correlation

	BI	GDP	CAR	Lerner	NPL	ROA_vol	LC	Ln Asset	LDR
BI	1								
GDP	-0.21	1							
CAR	0.07	-0.04	1						
Lerner	-0.13	0.02	-0.23	1					
NPL	0.16	-0.06	-0.08	0.03	1				
ROA_vol	0.23	-0.25	-0.19	0.41	0.15	1			
LC	-0.10	-0.04	-0.43	0.07	0.06	0.06	1		
Ln Asset	-0.18	0.04	-0.30	0.91	0.05	0.36	0.15	1	
LDR	-0.07	-0.03	0.01	-0.02	0.02	-0.13	0.33	0.00	1

Table 5 shows coefficients of correlation between variables. Relatively low to moderate values of coefficient (less than 0.50) confirming that in general there is no serious *multicollinearity* problem between explanatory variables, except for between Ln Asset and Lerner that exhibits very strong correlation with coefficient of 0.91. This indicates that larger banks are strongly associated with higher market power than the small ones. Due to this very strong correlation with Lerner that may lead to serious multicollinearity problem, I exclude Ln Asset and utilize Lerner as a proxy to control for banks' size when modelling LC in baseline and 2nd stage of IV regressions.

IV.2 Regression Results

IV.2.1 Overall sample

The baseline regression in general shows that market power and macro prudential regulation of reserve requirement linked to loan to deposit ratio have significant impact on liquidity creation. The positive sign of coefficient of Lerner indicates that banks with larger market power are able to create more liquidity. Thus, higher market power leads to higher liquidity creation. The positive sign of coefficient of d_RRLDR confirms that implementation of this macro prudential measure significantly increases banking liquidity creation.

The first stage of IV regression shows strong significant impact of transparency of prime lending rate regulation d_PLR on Lerner, confirming that this regulation is a valid instrument variable for Lerner and is effectively promote stronger price competition among banks. However, the transparency of prime lending rate regulation has an unintended consequence on liquidity creation. Consistent with the baseline regression, the 2nd stage of IV regression shows positive and significant relationship between Lerner and LC, implying that stronger banking competition due to the transparency of prime lending rate actually contains liquidity creation. Nevertheless, lower liquidity creation is not always necessarily bad, but it also implies a more stable banking system in terms of lower liquidity risk. This is inline with competition-fragility theorem that points out higher competition reduces banks' profits which act as buffer against adverse shocks thus encourages banks to behave more prudently by limiting both the volume of loans granted and the volume of deposit issued. This risk cautious behaviour of Indonesia banking can be attributed to the country experiences on several major crises that bring negative consequences on the industry, including the 1997/98 East Asia crisis, 2005 crises, and 2008/09 global financial crisis. The economic impact of stronger competition on reducing liquidity creation is somewhat small as shown in table 8. The 17.82% improvement (or equal to around 0.4 of negative standard deviation shock) of Lerner index due to implementation of transparency in prime lending rate leads to only 1.56% decrease

of liquidity creation as compared to average liquidity creation ratio at the level of around 40%.

The implementation of macro prudential regulation of reserve requirement linked to loan to deposit ratio achieves the intended objective with larger economic impact. It boosts liquidity creation up by around 2.62% to 5.39% higher than before the implementation. Therefore, macro prudential regulation of reserve requirement linked to loan to deposit ratio is effective to foster bank liquidity creation while transparency in prime lending rate is effective to improve competitiveness among banks with small impact on lower liquidity creation.

Both baseline and IV regressions confirm potential trade off between maintaining higher capital ratio for a more resilient banking system and creating lower liquidity to the economy. This finding is in line with the *financial fragility structure* theorem proposed by Berger & Bouwman (2009). The negative and significant signs of ROA_vol and NPL indicate that higher risks in terms of higher uncertainty in earning and credit risk encourage banks to act more competitive.

The outputs also indicate that expansionary monetary policy in terms of lower interest rate BI is effective to foster liquidity creation. The impact is somewhat moderate as one standard deviation of negative shock of BI will lead to around 1% increase of liquidity creation in the next three quarter.

Table 6. Estimation results of baseline and IV regressions for all banks

Independent Variables	Dependent Var: LC	Independent Variables	Dependent Var: Lerner	Dependent Var: LC
	Baseline		1st stage	2nd stage
Lerner (-1)	0.020 [2.10]**	d_PLR	-17.815 [-2.05]**	
CAR (-1)	-0.519 [-14.14]***	Lerner hat (-1)		0.079 [1.94]*
ROA_vol (-3)	0.041 [0.84]	Ln Asset (-4)	21.057 [3.92]***	
NPL (-2)	-0.033 [-0.21]	CAR (-1)	-0.189 [-0.77]	-0.504 [-14.69]***
GDP (-3)	0.216 [1.52]	ROA_vol (-2)	-1.141 [-4.85]***	
BI (-3)	-0.455 [-4.05]***	ROA_vol (-3)		0.087 [1.36]
d_RRLDR	5.386 [5.05]***	NPL (-2)	-0.519 [-1.95]*	0.037 [0.30]
constant	51.848 [32.31]***	GDP (-3)	-0.930 [-1.70]*	0.037 [0.28]
Time Fixed Effect	yes	BI (-2)	2.337 [5.30]***	
Bank Fixed Effect	yes	BI (-3)		-0.583 [-4.12]***
R-squared	0.22	d_RRLDR		2.618 [2.26]**
No. Obs	3819	constant	47.874 [5.63]***	51.263 [19.97]***
		Time Fixed Effect	yes	yes
		Bank Fixed Effect	yes	yes
		R-squared	0.61	0.20
		No. Obs	3724	3612

Note: Robust t-statistics are reported in in parantheses, with *, **, *** indicate statistical significance at the level of 10%, 5%, and 1%, respectively

Table 7. Economic impact of competition and macroprudential regulation on liquidity creation

PLR on Lerner	1 sd negative shock of Lerner on LC		RRLDR on LC	
IV reg	IV reg	baseline reg	IV reg	baseline reg
-17.81	-3.91	-0.93	2.62	5.39

IV.2.2 Large banks vs small banks

The estimation results of baseline and IV regressions that utilize sample of large and small banks separately are summarized in Table 8 and Table 9, respectively. It is interesting to know that while transparency of prime lending rate indeed improves significantly price competitiveness among large banks, it does not significantly lead

to higher liquidity creation. This regulation does not promote price competitiveness among small banks and there is weak indication that more market power will encourage small banks to create more liquidity.

The macro prudential measure of reserve requirement linked to loan to deposit ratio improves significantly the liquidity creation of small banks as indicated by positive and significant coefficient of d_RRLDR, but not in the case of large banks. As small banks contribute to around one third of total intermediated industry loan, this may justify the relatively moderate economic impact (boost up LC by around 2.62%-5.39% vis a vis average level of around 40%) of this regulation on banking aggregate liquidity creation as revealed in the regression results in Table 6.

Table 8. Estimation results of baseline regressions: large banks and small banks

Independent Variables	Dependent Var: LC	
	Baseline 20 large banks	Baseline small banks
Lerner (-1)	0.108 [0.24]	0.013 [1.33]
CAR (-1)	-0.213 [-1.66]	-0.530 [-14.29]***
ROA_vol (-3)	0.081 [0.89]	0.201 [3.28]***
NPL (-2)	-0.214 [-1.47]	-0.231 [-0.18]
GDP (-3)	0.558 [2.08]*	0.194 [1.15]
BI (-3)	-0.465 [-2.35]**	-0.748 [-5.96]***
d_RRLDR	2.172 [1.18]	8.651 [5.35]***
constant	37.277 [0.84]	51.350 [28.95]***
Time Fixed Effect	yes	yes
Bank Fixed Effect	yes	yes
R-squared	0.08	0.24
No. Obs	820	2999

Note: Robust t-statistics are reported in in parantheses, with *, **, *** indicate statistical significance at the level of 10%, 5%, and 1%, respectively

There is also trade off between maintaining higher capital ratio for a more resilient banking system and creating more liquidity to the economy for both large and small banks.

It is interesting to know that the 1st stage of IV regressions in Table 9 indicate high uncertainty in profit driving significantly price competitiveness among large banks while higher credit risk level NPL to some extent promote stronger competitiveness among small banks.

Table 9. Estimation results of IV regressions: large banks and small banks

Independent Variables	Dependent Var: Lerner	Dependent Var: LC	Dependent Var: Lerner	Dependent Var: LC
	1st stage 20 large banks	2nd stage 20 large banks	1st stage small banks	2nd stage small banks
d_PLR	-5.958 [-3.33]***		-0.388 [-0.04]	
Lerner hat (-1)		-0.076 [-0.17]		0.089 [1.79]*
Ln Asset (-4)	5.223 [4.47]***		22.974 [3.99]***	
CAR (-1)	-0.044 [-1.86]*	-0.223 [-1.77]*	-0.147 [-0.58]	-0.510 [-14.64]***
ROA_vol (-2)	-0.213 [-3.70]***		-0.223 [-1.01]	
ROA_vol (-3)		0.060 [0.53]		0.186 [2.97]***
NPL (-2)	-0.085 [-1.59]	-0.137 [-1.05]	-0.572 [-1.85]*	0.064 [0.43]
GDP (-3)	0.015 [0.17]	0.385 [1.48]	0.117 [0.17]	0.009 [0.06]
BI (-2)	0.305 [3.59]***		0.767 [1.46]	
BI (-3)		-0.481 [-2.34]**		-0.799 [-5.56]***
d_RRLDR		1.630 [0.83]		4.123 [1.90]*
constant	81.862 [20.55]***	57.749 [1.36]	37.739 [4.39]***	51.233 [22.97]***
Time Fixed Effect	yes	yes	yes	yes
Bank Fixed Effect	yes	yes	yes	yes
R-squared	0.55	0.07	0.66	0.20
No. Obs	800	780	2924	2832

Note: Robust t-statistics are reported in parantheses, with *, **, *** indicate statistical significance at the level of 10%, 5%, and 1%, respectively

IV.3 Robustness check

IV.3.1 Placebo test

I assume transparency of prime lending rate and macro prudential regulation of reserve requirement linked to loan to deposit ratio were issued around three to four years earlier (in 2007 or 2008). The coefficient regression for the two prudential initiatives should be indistinguishable from zero to ensure that other factors do not drive the significant impact we have before. Table 10 shows that all placebo

coefficients of d_RRLDR and d_PLR are insignificant, indicating that the significant effects are only observed when the two policies are actually implemented. Therefore, we can be more certain that our analysis based on the previous model specification is robust.

Table 10. Placebo baseline and IV regressions

Independent Variables	Dependent Var: LC	Independent Variables	Dependent Var: Lerner
	Baseline - Placebo		1st stage-Placebo
Lerner (-1)	0.019 [2.00]***	d_PLR	-1.365 [-0.87]
CAR (-1)	-0.520 [-14.25]***	Ln Asset (-4)	21.090 [3.92]***
ROA_vol (-3)	-0.013 [-0.16]	CAR (-1)	-0.189 [-0.77]
NPL (-2)	-0.038 [-0.25]	ROA_vol (-2)	-1.146 [-4.86]***
GDP (-3)	0.235 [1.64]	NPL (-2)	-0.520 [-1.95]*
BI (-3)	-0.537 [5.56]***	GDP (-3)	-1.012 [-1.68]*
d_RRLDR	-1.752 [-1.32]	BI (-2)	2.359 [5.38]***
constant	53.433 [27.17]***	constant	48.226 [5.53]***
Time Fixed Effect	yes	Time Fixed Effect	yes
Bank Fixed Effect	yes	Bank Fixed Effect	yes
R-squared	0.22	R-squared	0.61
No. Obs	3819	No. Obs	3724

Note: Robust t-statistics are reported in parantheses, with *, **, *** indicate statistical significance at the level of 10%, 5%, and 1%, respectively

IV.3.2 Liquidity creation measure without equity component

By construction, the LC measure contains equity component. The equity component is also a part of the capital adequacy ratio CAR which serves as a control variable in both baseline and IV regression. To ensure that the main results are robust to this potential overlap problem, I exclude equity component from the initial LC measure and re do both baseline and IV regression by utilizing the new liquidity creation indicator LC_{new} . The output of the regressions is summarized in Table 11.

Table 11. Baseline and IV regressions without equity component in LC

Independent Variables	Dependent Var: newLC	Independent Variables	Dependent Var: Lerner	Dependent Var: LCnew
	Baseline		1st stage	2nd stage
Lerner (-1)	0.007 [0.72]	d_PLR	-17.815 [-2.05]**	
CAR (-2)	-0.359 [-9.90]***	Lerner hat (-1)		0.071 [1.85]*
ROA_vol (-3)	-0.005 [-0.12]	Ln Asset (-4)	21.057 [3.92]***	
NPL (-2)	-0.053 [-0.42]	CAR (-1)	-0.189 [-0.77]	
GDP (-3)	0.212 [1.59]	CAR (-2)		-0.312 [-8.92]***
BI (-3)	-0.436 [3.83]***	ROA_vol (-2)	-1.141 [-4.85]***	
d_RRLDR	6.125 [5.64]***	ROA_vol (-3)		0.059 [0.92]
constant	55.900 [31.01]***	NPL (-2)	-0.519 [-1.95]*	0.044 [0.48]
		GDP (-3)	-0.930 [-1.70]*	0.095 [0.71]
Time Fixed Effect	yes	BI (-2)	2.337 [5.30]***	
Bank Fixed Effect	yes	BI (-3)		-0.599 [-4.25]***
R-squared	0.10	d_RRLDR		3.114 [2.71]***
No. Obs	3821	constant	47.874 [5.63]***	54.206 [20.23]***
		Time Fixed Effect	yes	yes
		Bank Fixed Effect	yes	yes
		R-squared	0.61	0.08
		No. Obs	3724	3624

Note: Robust t-statistics are reported in in parantheses, with *, **, *** indicate statistical significance at the level of 10%, 5%, and 1%, respectively

In general, the estimation results suggest relatively similar conclusion as do by the previous main analyses. Regulation of transparency in prime lending rate is effective to enhance price competition among banks that leads to lower liquidity creation, while macro prudential measure of reserve requirement linked to loan to deposit ratio is effective to improve banking liquidity creation. Therefore, the main conclusion is robust to LC composition issue.

IV.3.3 Loan to deposit ratio (LDR) as another proxy for liquidity creation

As macroprudential policy in reserve requirement linked to loan to deposit ratio is directly aimed at improving banks liquidity creation with LDR as the benchmark, I consider LDR as another proxy for liquidity creation. Even though LDR is less

comprehensive in measuring banks liquidity creation than LC, it represents both banks' intermediation role in collecting and channeling funds as well as potential liquidity risk. Higher LDR can also be associated for higher liquidity creation and liquidity risk. Table 5 indicates moderate positive correlation (0.33) between LDR and LC, supporting it as another proxy for liquidity creation.

The estimation results utilizing LDR as dependent variable are displayed in Table 12.

Table 12. Baseline and IV regression for LDR as a proxy for liquidity creation

Independent Variables	Dependent Var: LDR	Independent Variables	Dependent Var: Lerner	Dependent Var: LDR
	Baseline		1st stage	2nd stage
Lerner (-1)	-0.031 [-1.45]	d_PLR	-17.815 [-2.05]**	
CAR (-1)	-0.141 [-2.59]**	Lerner hat (-1)		0.017 [0.22]
ROA_vol (-3)	0.059 [0.60]	Ln Asset (-4)	21.057 [3.92]***	
NPL (-2)	-0.184 [-0.85]	CAR (-1)	-0.189 [-0.77]	-0.162 [-2.46]**
GDP (-3)	-0.319 [0.30]	ROA_vol (-2)	-1.141 [-4.85]***	
BI (-3)	-1.014 [-3.84]***	ROA_vol (-3)		0.087 [0.77]
d_RRLDR	15.962 [7.34]***	NPL (-2)	-0.519 [-1.95]*	-0.174 [-0.84]
constant	83.153 [27.62]***	GDP (-3)	-0.930 [-1.70]*	-0.508 [-1.60]
Time Fixed Effect	yes	BI (-2)	2.337 [5.30]***	
Bank Fixed Effect	yes	BI (-3)		-1.067 [-3.55]***
R-squared	0.05	d_RRLDR		11.102 [4.44]***
No. Obs	3480	constant	47.874 [5.63]***	85.772 [15.95]***
		Time Fixed Effect	yes	yes
		Bank Fixed Effect	yes	yes
		R-squared	0.61	0.03
		No. Obs	3724	3299

Note: Robust t-statistics are reported in in parantheses, with *, **, *** indicate statistical significance at the level of 10%, 5%, and 1%, respectively

Table 12 confirms consistent results with previous analyses that implementation of macroprudential regulation reserve requirement linked to loan to deposit ratio is effective to increases banks liquidity creation as indicated by positive and significant sign of d_RRLDR in both baseline and IV regressions. However, competition does not play significant role in affecting banks liquidity creation as confirmed by non significant coefficient of Lerner in baseline and Lerner_hat in IV regression.

IV.4 Alternative explanation and possible extension

That stronger competition contributes to lower liquidity creation in main analysis deserves for further clarification.

So far, I evaluate the role of this stronger competition on banks liquidity creation from the volume dimension of banks intermediation. However, one could also evaluate the effectiveness of this regulation from the price dimension. Following the result in the first stage of IV analysis that confirms the regulation is effective to foster stronger price competition in banking, we could examine the hypotheses if this stronger competition leads to significantly lower banks loans interest rates in second stage, which is the objective of the regulation of prime lending rate.

Table 13 displays the estimation results for IV analysis. All estimated coefficients in first stage are just the same as those in main analysis in Table 6. In the second stage, I regress loans interest rates on predicted Lerner from the first stage including same control variables.

The negative and significance coefficient of $\widehat{\text{Lerner}}$ indicates that stronger competition due to implementation of transparency in prime lending rate induce banks to charge customers higher loan interest rates. This may be caused by banks attempt to maintain their profit level as stronger competition seems to erode banks profit, which is inline with the competition-fragility theorem. Moreover, the significance and positive value of NPL confirms the tendency of risk averse behaviour. Higher credit risk encourages banks to increase their loan interest rates.

If we assume that higher loans rates will reduce demand for loans leading to lower liquidity creation, *ceteris paribus*, this will support the conclusion drawn from the main analysis in Table 6 that stronger competition actually leads to lower liquidity creation that may be attributed to banking risk averse behaviour.

I leave detail of this empirical part to evaluate the hypotheses above on the link between loan interest rates and liquidity creation as future work.

Table 13. IV regression for Loan interest rate

Independent Variables	Dependent Var: Lerner	Dependent Var: Loan interest rate
	1st stage	2nd stage
d_PLR	-17.815 [-2.05]**	
Lerner hat (-4)		-0.046 [-1.94]*
Ln Asset (-4)	21.057 [3.92]***	
CAR (-1)	-0.189 [-0.77]	
CAR (-3)		-0.015 [-0.60]
ROA_vol (-1)		0.098 [0.79]
ROA_vol (-2)	-1.141 [-4.85]***	
NPL (-1)		0.185 [2.39]**
NPL (-2)	-0.519 [-1.95]*	
GDP (-3)	-0.930 [-1.70]*	-0.114 [-0.59]
BI (-1)		0.593 [3.35]***
BI (-2)	2.337 [5.30]***	
d_RRLDR		-0.111 [-0.05]
constant	47.874 [5.63]***	12.640 [3.04]***
Time Fixed Effect	yes	yes
Bank Fixed Effect	yes	yes
R-squared	0.61	0.25
No. Obs	3724	3323

Note: Robust t-statistics are reported in in parantheses, with *, **, *** indicate statistical significance at the level of 10%, 5%, and 1%, respectively

V. Conclusion and policy implication

This study provides insights into the role of competition and effectiveness of prudential regulations aimed at improving banking liquidity creation post global financial crises 2008/09 by utilizing a novel measure of bank liquidity creation.

In general the results confirm that macro prudential regulation of reserve requirement linked to loan to deposit ratio is effective enough to increase banks liquidity creation. Regulation of the transparency in prime lending rate is also effective to foster stronger price competition among banks.

Even though it might need further clarification, there are some evidences that stronger competition leads to lower liquidity creation as banks increase their loans interest rates. This finding is in line with the competition-fragility theorm where

higher competition leads banks to be more cautious on risk thus limiting liquidity creation. Risk cautious behavior of Indonesia banking can be attributed to the country experiences on several major crises that bring negative consequences on the industry, including the 1997/98 East Asia crisis, 2005 domestic crises due to oil price liberalization, and 2008/09 global financial crisis. Nevertheless, lower liquidity creation is not always necessarily bad, as it also implies a more stable banking system in terms of lower liquidity risk.

The risk averse behaviour is more pronounce among small banks. Therefore, consolidation such as merger among small banks may be a good idea to get greater market power that enables them to create more liquidity. This policy option however needs to weigh in between the efficiency gains in terms of more favourable loan prices for costumers and the possible downside risks of consolidation, including too-big-to fail institutions.

All in all, on net, implementing both of these regulations encourage banks to create more liquidity and promote prudent behaviour, therefore enhancing banking stability.

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